Practical No. 12 – Hypothesis Testing for One Population Proportion and Difference of Population Proportions

Blue = given data (inputs). Black = formulas. Each test is solved by the critical-value method and the p-value method; both must agree.

Ex.1 Bluetooth transceivers – one-proportion z-test (right-tailed)

H0: p = 0.70 vs H1: p > 0.70

Sample size n	60	
Number of successes x	48	
Hypothesised proportion p0	0.70	
Significance level α	0.05	
Sample proportion p̂	0.8000	*p̂ = x/n*
Standard error (under H0)	0.05916	*√(p0(1−p0)/n)*
Test statistic z	**1.6903**	*z = (p̂ − p0)/SE*
Critical value z(α)	1.6449	*NORMSINV(1−α)*
p-value (right tail)	0.04548	*1−Φ(z)*
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Reject H0 – the pass proportion is significantly greater than 0.70.

Ex.2 Airline lost luggage – one-proportion z-test (right-tailed)

H0: p = 0.06 vs H1: p > 0.06

Sample size n	200	
Number of successes x	17	
Hypothesised proportion p0	0.06	
Significance level α	0.05	
Sample proportion p̂	0.0850	*p̂ = x/n*
Standard error (under H0)	0.01679	*√(p0(1−p0)/n)*
Test statistic z	**1.4887**	*z = (p̂ − p0)/SE*
Critical value z(α)	1.6449	*NORMSINV(1−α)*
p-value (right tail)	0.06828	*1−Φ(z)*
Decision (critical value method)	**Fail to reject H0**	
Decision (p-value method)	**Fail to reject H0**	

Conclusion: Fail to reject H0 – not enough evidence that more than 6% is never found.

Ex.3 St. John's Wort vs placebo – two-proportion z-test (right-tailed)

H0: p1 = p2 vs H1: p1 > p2 (1 = St. John's Wort, 2 = placebo; 200 patients split equally → 100 each)

n1 (St. John's Wort)	100	
x1 – successes (St. John's Wort)	27	
n2 (Placebo)	100	
x2 – successes (Placebo)	19	
Significance level α	0.05	
Sample proportion p̂1	0.2700	*x1/n1*
Sample proportion p̂2	0.1900	*x2/n2*
Pooled proportion p̄	0.2300	*(x1+x2)/(n1+n2)*
Standard error	0.05951	*√(p̄(1−p̄)(1/n1+1/n2))*
Test statistic z	**1.3442**	*z = (p̂1 − p̂2)/SE*
Critical value z(α)	1.6449	*NORMSINV(1−α)*
p-value (right tail)	0.08944	*1−Φ(z)*
Decision (critical value method)	**Fail to reject H0**	
Decision (p-value method)	**Fail to reject H0**	

Conclusion: Insufficient evidence that St. John's Wort is effective in treating major depression.

Ex.4 A/B testing of web page – two-proportion z-test (right-tailed)

H0: p1 = p2 vs H1: p1 > p2 (1 = modified page, 2 = original page)

n1 (Modified page)	2297	
x1 – successes (Modified page)	107	
n2 (Original page)	2841	
x2 – successes (Original page)	77	
Significance level α	0.05	
Sample proportion p̂1	0.0466	*x1/n1*
Sample proportion p̂2	0.0271	*x2/n2*
Pooled proportion p̄	0.0358	*(x1+x2)/(n1+n2)*
Standard error	0.00521	*√(p̄(1−p̄)(1/n1+1/n2))*
Test statistic z	**3.7360**	*z = (p̂1 − p̂2)/SE*
Critical value z(α)	1.6449	*NORMSINV(1−α)*
p-value (right tail)	0.00009	*1−Φ(z)*
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Strong evidence that the proportion of purchasers is higher for the modified page.